

## **PYTHON-TEST-2**

MCQ

(½ X32)=16M

**Q1. What is the primary purpose of the if statement in Python?**

- a) To execute a block of code based on a condition      b) To perform mathematical operations  
c) To repeat a block of code      d) To define a function

**Q2. What will be the output of the following code snippet?**

```
x = 10
```

```
if x > 5:
```

```
    print("x is greater than 5")
```

- a) x is greater than 5      b) x is less than 5      c) x is equal to 5      d) No output

**Q3. Which keyword is used to execute a block of code if the condition in the if statement is false?**

- a) else      b) elif      c) while      d) for

**Q4. What is the purpose of the elif statement in Python?**

- a) To execute a block of code if the previous conditions are false      b) To define a loop  
c) To perform arithmetic operations      d) To exit from a loop

**Q5. What will be the output of the following code snippet?**

```
x = 5
```

```
if x < 3:
```

```
    print("x is less than 3")
```

```
elif x == 3:
```

```
    print("x is equal to 3")
```

```
else:
```

```
    print("x is greater than 3")
```

- a) x is less than 3      b) x is equal to 3      c) x is greater than 3      d) No output

**Q6. How can you execute multiple statements under a single if block in Python?**

- a) Separate statements with a semicolon      b) Indent the statements to the same level  
c) Use the and keyword between statements      d) Use the elif keyword

**Q7. What will the following code snippet print?**

```
x = 10
```

```
if x < 5:
```

```
    print("x is less than 5")
```

```
elif x > 15:
```

```
    print("x is greater than 15")
```

```
else:
```

```
    print("x is between 5 and 15")
```

- a) x is less than 5      b) x is greater than 15      c) x is between 5 and 15      d) No output

**Q8. What is the purpose of the while loop in Python?**

- a) To execute a block of code repeatedly until a condition is false  
b) To execute a block of code a fixed number of times  
c) To define a function      d) To iterate over items in a sequence

**Q9. What is the syntax for a while loop in Python?**

- a) while condition:      b) while condition():      c) while (condition):      d) while loop condition:

**Q10. How can you exit a loop prematurely in Python?**

- a) Using the break statement      b) Using the continue statement  
c) Using the pass statement      d) Using the exit function

**Q11. What is the purpose of the for loop in Python?**

- a) To execute a block of code repeatedly until a condition is false  
b) To iterate over items in a sequence      c) To execute a block of code a fixed number of times  
d) To define a function

**Q12. What is the syntax for a for loop in Python?**

- a) for item in sequence:  
b) for item in range(n):  
c) for index in range(len(sequence)):  
d) All of the above

**Q13. How can you skip the current iteration of a loop and continue with the next iteration?**

- a) Using the skip statement      b) Using the pass statement
- c) Using the break statement      d) Using the continue statement

**Q14. What is the purpose of the range() function in Python?**

- a) To generate a sequence of numbers      b) To iterate over items in a sequence
- c) To define a function      d) To execute a block of code repeatedly until a condition is false

**Q15. What will the following code snippet print?**

```
for i in range(3):
```

```
    print(i)
```

- a) 0 1 2      b) 1 2 3      c) 2 1 0      d) 3 2 1 0

**Q16. What will be the output of the following code?**

```
for i in range(1, 6):
```

```
    if i == 3:
```

```
        continue
```

```
    print(i)
```

- a) 1 2      b) 1 2 3      c) 1 2 4 5      d) 1 2 4

**Q17. What will be the output of the following code?**

```
for i in range(3):
```

```
    for j in range(3):
```

```
        print(i + j, end=' ')
```

```
    print()
```

- a) 0 1 2

1 2 3

2 3 4

- b) 0 1 2

1 2 3

2 3 4

3 4 5

c) 0 1 2

1 2 3

2 3 4

4 5 6

d) 0 1 2

2 3 4

4 5 6

**Q18. What will be printed by the following code?**

```
num = 5
```

```
while num > 0:
```

```
    print(num)
```

```
    num -= 1
```

```
    if num == 3:
```

```
        break
```

```
else:
```

```
    print("Done")
```

a) 5 4 3 2 1

b) 5 4

c) Done

d) 5 4 3

**Q19. What is the output of the following code?**

```
x = 10
```

```
if x > 5:
```

```
    print("A")
```

```
elif x > 7:
```

```
    print("B")
```

```
else:
```

```
    print("C")
```

- a) A      b) B      c) C      d) A and B

**Q20. What will be the output of the following code?**

```
x = 5
```

```
while x > 0:
```

```
    print(x, end=" ")
```

```
    x -= 2
```

```
    if x == 1:
```

```
        break
```

```
else:
```

```
    print("Done")
```

- a) 5 3 1      b) 5 3      c) 5 3 Done      d) 5 3 1 Done

**Q21. What is the output of the following code?**

```
for i in range(5):
```

```
    if i == 2:
```

```
        continue
```

```
    print(i, end=" ")
```

- a) 0 1 3 4      b) 0 1 2 3 4      c) 0 1 3 4 5      d) 0 1 2 3 4 5

**Q22. What is the output of the following code?**

```
num = 0
```

```
while num < 5:
```

```
    print(num)
```

```
    num += 1
```

```
else:
```

```
    print("Loop completed.")
```

- a) 0 1 2 3 4      b) 0 1 2 3 4 Loop completed.      c) Loop completed.      d) This code will result in an error.

**Q23. What will be the output of the following code?**

```
x = 10
```

```
if x > 5:
```

```
    print("Greater than 5")
if x > 8:
    print("Greater than 8")
if x > 12:
    print("Greater than 12")
else:
    print("Equal or less than 12")
```

a) Greater than 5

Greater than 8

Equal or less than 12

b) Greater than 5

Greater than 8

Greater than 12

Equal or less than 12

c) Greater than 5

Greater than 8

d) Equal or less than 12

**Q24. What will be the output of the following code?**

```
for i in range(3):
    for j in range(3):
        print(i * j, end=' ')
    print()
```

a) 0 0 0

0 1 2

0 2 4

b) 0 1 2

0 2 4

0 3 6

c) 0 0 0

0 1 2

0 2 4

0 3 6

**d) 0 0 0**

0 1 2

0 1 2

**Q25. What will be the output of the following code?**

```
num = 10
```

```
while num > 0:
```

```
    print(num)
```

```
    num -= 3
```

```
else:
```

```
    print("Loop completed.")
```

a) 10 7 4 1 Loop completed.      b) 10 7 4 1      c) Loop completed.      d) This code will result in an error.

**Q26. What will the following code snippet print?**

```
for i in range(3):
```

```
    pass
```

```
print(i)
```

a) 0 1 2      b) 1 2 3      c) 2 1 0      d) 3 2 1 0

**Q27. What is the purpose of the pass statement in Python?**

a) To exit from a loop prematurely      b) To skip the current iteration of a loop

c) To execute a block of code if a condition is false      d) To do nothing and act as a placeholder

**Q28. What is the purpose of the else block in a loop in Python?**

a) To execute if the loop encounters an error

b) To execute if the loop completes without encountering a break statement

c) To execute if the loop encounters a continue statement

d) To execute if the loop encounters a pass statement

**Q29. What is the output of the following code?**

```
x = 10
```

if x > 5:

    print("Hello")

elif x > 8:

    print("Hi")

else:

    print("Hey")

a) Hey            b) Hi            c) Hello            d) Hello, Hi

**Q30. What is the output of the following code?**

for i in range(1, 6):

    if i % 2 == 0:

        print(i)

        continue

    print("\*")

a) \n2\n\n4\n\*      b) \n2\n\n4\n\*\n      c) \n2\n\n\*\n4\n      d) \n\n2\n\*\n4\n\n

**Q31. What will be the output of the following code?**

x = 5

while x > 0:

    print(x)

    x -= 1

else:

    print("Done")

a) 5\n4\n3\n2\n1\nDone      b) Done\n5\n4\n3\n2\n1      c) 5\n4\n3\n2\n1      d) Done

**Q32. What will be the output of the following code?**

for i in range(1, 6):

    if i == 3:

        break

    print(i)

else:

    print("Loop completed.")



- a) 1\n2      b) 1\n2\n3\n4\n5\nLoop completed.      c) 1\n2\n3\n4\n5      d) 1\n2\n3

**2 MARKS**

(2X5)=10M

1. what is python and Difference between .py file and .pyc file ?
- 2.what is dynamic type with example?
  - i.what is PEP8?
  - ii.what is interpreter ?
  - iii.how do you comment in python?
- 3.what are the datatypes in python?
  - i.what is python operators and types of operators?
- 4.what is flow control statements in python ?Difference b\w for and while loop?
- 5.Difference between python 2 and python 3?

**4MARKS**

1. Write the program to print right angular pattern.

**20MARKS**

1. Write the proگرام ?

**1.**  
 E D C B A  
 D C B A  
 C B A  
 B A  
 A  
**2.**  
 A 1  
 B 2 C 3  
 D 4 E 5 F 6  
 G 7 H 8 I 9 J 10  
**3.**  
 G 1 E 4 C 9 A  
 F 1 D 4 B 9  
 E 1 C 4 A  
 D 1 B 4  
 C 1 A  
 B 1  
 A  
**4.**  
 I 3 G 4 E 5 C  
 H 3 F 4 D C  
 G 3 E 4 C  
 E 3 D 4  
 E 3 C  
 D 3  
 C  
**5.**  
 A  
 1 3  
 C E G  
 5 7 9 11  
 I K M O Q

**1.)**  
**EDCBA**  
**DCBA**  
**CBA**  
**BA**  
**A**

```

1. for i in range(5, 0, -1):
    for j in range(i, 0, -1):
        print(chr(64 + j), end="")
    print()

```

**Row (i) j Loop Values Characters Printed**

|   |               |       |
|---|---------------|-------|
| 5 | 5, 4, 3, 2, 1 | EDCBA |
| 4 | 4, 3, 2, 1    | DCBA  |
| 3 | 3, 2, 1       | CBA   |
| 2 | 2, 1          | BA    |
| 1 | 1             | A     |

```

2. A1
   B2C3
   D4E5F6
   G7H8I9J10

```

Ans:

```

char = 65 # ASCII value of 'A'
num = 1
for i in range(1, 5):
    for j in range(i):
        print(f"{chr(char)}{num}", end="")
        char += 1
        num += 1
    print()

```

```

3.
G1E4C9A
F1D4B9

```

E1C4A  
D1B4  
C1A  
B1  
A

Ans:

```
letters = ['G', 'E', 'C', 'A']
nums = [1, 4, 9]
for i in range(7, 0, -1):
    for j in range(0, i, 2):
        if j // 2 < len(letters):
            print(letters[j // 2], end="")
        if j // 2 < len(nums):
            print(nums[j // 2], end="")
    print()
```

4.  
I3G4E5C  
H3F4DC  
G3E4C  
E3D4  
E3C  
D3  
C

Ans:

```
chars = ["I", "H", "G", "E", "D", "C"]
nums = [3, 4, 5]
for i in range(len(chars)):
    for j in range(0, len(chars) - i, 2):
        print(chars[j], end="")
        if j // 2 < len(nums):
            print(nums[j // 2], end="")
    print()
```

5.

A

13

CEG

57 9 11

IKMOQ

Ans:

```
print("A")
```

```
print("13")
```

```
print("CEG")
```

```
print("57 9 11")
```

```
print("IKMOQ")
```

4mark

EDCBA

DCBA

CBA

BA

A

```
for i in range(5, 0, -1):
```

```
    for j in range(i, 0, -1):
```

```
        print(chr(64 + j), end="")
```

print()

# functions in python

12-09-2024

## 1.create a function called add(),sub(),mul(),div()

**-And get the input for a and b inside every function print the result.**

2.Get a integer number from user and pass it to the function called findevenorodd().

Let the function print whether the number is even or odd.

3.Get a integer number from user and pass it to the function called findpassorfail().Let the function print whether pass or fail.(<35---fail)

4.Get input for a and b and pass it to the function called printrange().Let the function print numbers from a to b